Making waves

Musical chords are played on a guitar by strumming up to 6 strings simultaneously with one hand, while using fingers on the other hand t shorten the strings by different amounts to modify the notes each string plays. An E-major chord is played by by strumming the 6 strings v strings reverberate with the following fundamental frequencies:

Note	Frequency
1: E	330 Hz
2: B	247 Hz
3: G#	208 Hz
4: E	165 Hz
5: B	123 Hz
6: E	82 Hz

Plucking a guitar string also generates higher frequency harmonics in addition to the fundamental frequency that give it a full, rich sound. sounds, however, in this exercise, we will simulate a simple "pure tone" guitar.

Your task is to write a script that creates a sound signal by combining the 6 pure tones.

- create a row vector **t** that spans [0, 3) seconds using a sampling frequency, **Fs** = 44100 Hz
- create an output signal **y** that is the sum of cosines (amplitude = 1, phase = 0) with the frequencies provided in the table.

Script

Save Reset MATLAB Documentation (https://www.mathworks.com/help/)

```
% Define T, Fs, and f (vector of sound frequencies)
T = 3;
Fs = 44100;
f = [330, 247, 208, 165, 123, 82];
% Compute time vector, t
t = [0:(1/Fs):T];
if t(end) == T
    t = t(1:(end-1));
end
% Use a for-loop to construct y as a sum of cosines
y = zeros(size(t));
%size(y)
for i=1:numel(f)
y = y + cos(2*pi*f(i)*t);
end
%plot(t(1:100), y(1:100));
%soundsc(y, Fs);
```

Run Script

Assessment: All Tests Passed

Submit

Is the time vector, t, correct?

Is the signal variable, y, correct?

Output

```
Code ran without output.
```